



Finding missing sides

GIVEN AN ANGLE AND THE LENGTH OF ONE SIDE OF A RIGHT TRIANGLE, THE OTHER SIDES CAN BE FOUND.

The trigonometry formulas can be used to find a length in a right triangle if one angle (other than the right angle) and one other side are known. Use a calculator to find the sine, cosine, or tangent of an angle.

Which formula?

The formula to use depends on what information is known. Choose the formula that contains the known side as well as the side that needs to be found. For example, use the sine formula if the length of the hypotenuse is known, one angle other than the right angle is known, and the length of the side opposite the given angle needs to be found.

SEE ALSO

◀ 160 What is trigonometry?

Finding missing angles

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Formulas

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▽ Calculator buttons

These calculator buttons recall the value of sine, cosine, and tangent for any value entered.



this is the
sine button

this is the
cosine button

this is the
tangent button

$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$$

△ The sine formula

This formula is used if one angle, and either the side opposite it or the hypotenuse are given.

$$\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$$

△ The cosine formula

Use this formula if one angle and either the side adjacent to it or the hypotenuse are known.

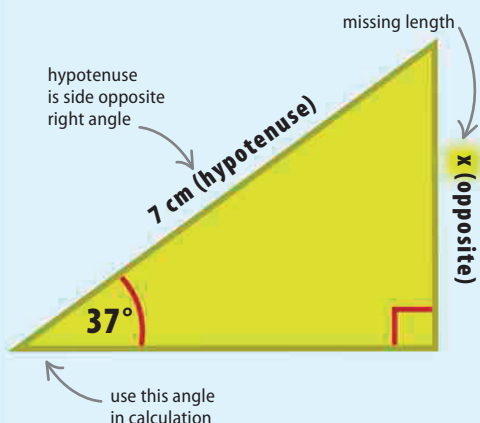
$$\tan A = \frac{\text{opposite}}{\text{adjacent}}$$

△ The tangent formula

This formula is used if one angle and either the side opposite it or adjacent to it are given.

Using the sine formula

In this right triangle, one angle other than the right angle is known, as is the length of the hypotenuse. The length of the side opposite the angle is missing and needs to be found.



Choose the right formula—because the hypotenuse is known and the value for the opposite side is what needs to be found, use the sine formula.

Substitute the known values into the sine formula.

Rearrange the formula to make the unknown (x) the subject by multiplying both sides by 7.

Use a calculator to find the value of $\sin 37^\circ$ —press the sin button then enter 37.

Round the answer to a suitable size.

$$\begin{aligned} \sin A &= \frac{\text{opposite}}{\text{hypotenuse}} \\ \sin 37^\circ &= \frac{x}{7} && \text{this side has also been multiplied by 7} \\ x &= \sin 37^\circ \times 7 && \text{show the unknown first} \\ x &= 0.6018 \times 7 && \text{this side has been multiplied by 7 to isolate x} \\ x &= 4.21 \text{ cm} && \text{this is value of } \sin 37^\circ \text{ to 4 decimal places} \\ &&& \text{answer is rounded to 2 decimal places} \end{aligned}$$